Selenium to Playwright Migration POC

Powered by IBM BOB | SauceDemo Automation Framework

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What We Will Cover

01 Project Overview

Selenium Java framework — goals and baseline

02 Selenium Baseline

21 scenarios, 5 modules, git tag selenium-baseline-v1

03 IBM BOB Migration

Converting Login + Products modules to Playwright TS

04 Side-by-Side Comparison

Java vs TypeScript — code, waits, assertions

05 Live Results + Key Takeaways

The Starting Point — Selenium Java Framework

Application Under Test

SauceDemo (saucedemo.com)
A standard e-commerce demo app used industry-wide for automation training

Framework Stack

Selenium 4.18 + Java 24
Cucumber BDD + TestNG
Maven build tool
ExtentReports + Log4j2

Test Coverage

5 modules: Login, Products Cart, Checkout, Logout
21 scenarios total
BDD .feature files

Goal: Reach 21/21 passing as a measurable Selenium baseline, then migrate each module to Playwright TypeScript using IBM BOB.

Bugs Fixed to Reach 21/21

Bug	Fix Applied
Stale driver in	Page objects resolved fresh per step via method — never captured as fields
Wrong CSS	Changed [data-test='product_sort_container'] to select.product_sort_container
Chrome	Added HashMap prefs + --disable-features=PasswordLeakDetection to ChromeOptions
Cart page not	goToCart() now waits for urlContains('cart.html') before returning
NoSuchElement	BasePage.click/type/getText upgraded to WebDriverWait(10s) explicit waits

Result: 21/21 passing git tag selenium-baseline-v1 Total run time: 3 min 21 sec

SECTION 02

IBM BOB Migration

Converting Selenium Java to Playwright TypeScript — module by module

Migration Approach — What IBM BOB Does

STEP 1
Read the source files

BOB reads LoginPage.java, LoginSteps.java and login.feature to understand every locator, method, step definition and assertion.

STEP 2
Map Java to TypeScript

By.id => #id
By.cssSelector => locator()
WebDriverWait => auto-wait
new Select() => selectOption()
Assert.assertEquals => expect().toBe()

STEP 3
Generate output files

login.page.ts — page object
login.spec.ts — test spec
All scenarios from .feature mapped 1-to-1 as test() blocks.

STEP 4
Run and verify

npx playwright test
Debug failures live
Fix and re-run until all pass.
Commit to git.

IBM BOB reads and writes code like a senior engineer — every change is grounded in the actual source files, never hallucinated.

Login Page Object — Java vs TypeScript

LoginPage.java (Selenium)

```java
private By username = By.id("user-name");
private By loginBtn = By.id("login-button");

public void login(String u, String p) {
  type(username, u);
  type(password, p);
  click(loginBtn);
  new WebDriverWait(driver, Duration.ofSeconds(15))
    .until(ExpectedConditions.or(
      urlContains("inventory.html"),
      visibilityOf(errorLocator)
    ));
}
```

By.id("x") = page.locator('#x')

login.page.ts (Playwright)

```typescript
private usernameInput = page.locator('#user-name');
private loginButton   = page.locator('#login-button');

async login(username: string, password: string) {
  await this.usernameInput.fill(username);
  await this.passwordInput.fill(password);
  await this.loginButton.click();
  await Promise.race([
    page.waitForURL('**/inventory.html'),
    this.errorMessage.waitFor({ state: 'visible' })
  ]);
}
```

WebDriverWait(15s) = Promise.race auto-wait

Feature File vs Playwright Spec

login.feature (Cucumber BDD)

```
Background:
  Given the user navigates to the login page

Scenario: Successful login
  When  the user logs in with
        username "standard_user"
        and password "secret_sauce"
  Then  the user should be
        on the products page

Scenario: Failed login
  When  the user logs in with
        username "standard_user"
        and password "wrong_password"
  Then  the user should see
        an error message "Epic sadface..."
```

login.spec.ts (Playwright)

```
test.beforeEach(async ({ page }) => {
  await loginPage.navigate(BASE_URL);
});

test('Successful login', async ({ page }) => {
  await loginPage.login(
    'standard_user', 'secret_sauce'
  );
  expect(await loginPage
    .isOnProductsPage()).toBeTruthy();
});

test('Failed login', async ({ page }) => {
  await loginPage.login(
    'standard_user', 'wrong_password'
  );
  expect(await loginPage
    .isOnProductsPage()).toBeFalsy();
});
```

Background = beforeEach() Scenario = test() Then (assertion) = expect()

Test Results — Both Frameworks

Selenium Java (selenium-baseline-v1)

```
ok  Login: Successful login                  PASS
ok  Login: Failed login - invalid password    PASS
ok  Login: Failed login - locked-out user     PASS
ok  Login: Failed login - blank username      PASS
ok  Login: Failed login - blank password      PASS
ok  Cart: Add single item                     PASS
ok  Cart: Add multiple items                  PASS
ok  Cart: Correct product name                PASS
ok  Checkout: Successful checkout             PASS
ok  Checkout: Total displayed                 PASS
ok  Checkout: Missing first name error        PASS
ok  Logout: Successfully                      PASS
ok  Products: Page displayed after login      PASS
ok  Products: Sort A to Z                     PASS
...

21 passed   Time: 3 min 21 sec
```

Playwright TypeScript (Login + Products)

```
ok  1  Login: Successful login                (3.7s)
ok  2  Login: Failed login invalid password    (3.3s)
ok  3  Login: Failed login locked-out user     (3.1s)
ok  4  Login: Failed login blank username      (3.1s)
ok  5  Login: Failed login blank password      (3.2s)
ok  6  Login: Multiple user types [standard]   (3.5s)
ok  7  Login: Multiple user types [problem]    (3.6s)
ok  8  Products: Page displayed after login    (3.2s)
ok  9  Products: Products list not empty       (3.1s)
ok 10  Products: Sort A to Z                   (3.1s)
ok 11  Products: Sort Z to A                   (3.2s)
ok 12  Products: Sort price low to high        (3.1s)
ok 13  Products: Sort price high to low        (3.4s)

13 passed   Time: 45 sec
```

Selenium vs Playwright — What Changed

Concept	Selenium Java	Playwright TypeScript
Locator	By.id / By.cssSelector	page.locator('#id')
Waiting	WebDriverWait(10s) explicit	Built-in auto-wait (no code)
Assertion	Assert.assertEquals(a, b)	expect(a).toBe(b)
Select dropdown	new Select(el).selectByVisibleText()	locator.selectOption({ label })
Driver lifecycle	DriverFactory + ThreadLocal	Built-in: test({ page }) fixture
Scenario Outline	Cucumber Examples table	for...of array over test cases

Speed: Selenium ~10 sec/test Playwright ~3.5 sec/test (3x faster)

Migration Roadmap

DONE

Login module
7 / 7 passing

Products module
6 / 6 passing

Baseline tagged
selenium-baseline-v1

NEXT

Cart module
4 scenarios

Checkout module
3 scenarios

Logout module
2 scenarios

FUTURE

CI / CD pipeline
Parallel execution
Cross-browser
Firefox + WebKit

To run today: cd playwright-login then npx playwright test

Thank You

Questions and discussion welcome

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Project: UIAutomationPOC / playwright-login Git tag: selenium-baseline-v1